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History and Current Capabilities in CFD++ for Overset Composite Grids

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Metacomp Technologies, Inc.

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Introduction

Having been involved in overset grids from the inception of the idea of an overset grid approach for implementation in flow solvers, a quick personal perspective on the early evolution of ideas through current day practical aspects is presented, the latter by discussing capabilities of CFD++. The ideas are illustrated with a variety of application scenarios from aerodynamics and hydrodynamics. Overset grids with zero-overlap are an important category of composite grids that will be discussed, along with the use of different physics/equations on different cell groups. Composite grids with relative motion and different node-point transformations are discussed along with the ability to couple grid motion with physics and multi-physics. Practical aspects including ease of use complete the presentation.

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Mid 70s onwards

- Development of body fitted grids
- Transformation of independent variables from Cartesian coordinate system to arbitrary curvilinear coordinate system
- Preservation of strong conservation law form in transformed coordinate systems
- Shock-fitting versus/and/or Shock-capturing
- Numerical mesh generation - elliptic mesh generation, for arbitrary geometries (Thompson, Thames, Mastin)
- Numerical treatment of transformations, Jacobians
- Development of Finite Difference Schemes based on the above – central differencing with smoothing operators

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Personal chronology 1976-1977

- Worked at NASA Ames Research Center during the summer months as part of Iowa State University program
- Worked under Paul Kutler, but developed rapport with Joe Steger – later connection to composite and overset grids (1982-1983)
- Development of elliptic grid generation code - 1976
- First “Full Navier-Stokes” codes in 2D and 3D using approximate-factorization, i.e., not using “Thin-Layer Navier-Stokes” equations - 1977
- Contributed to the evolution of “Parabolized Navier-Stokes” solvers using space marching - 1977

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Through Early 80s

- Structured grid codes (single block)
- Addition of Finite-Volume schemes
- Two types of solvers, all node based, both considered to satisfy numerical conservation laws, even for Finite-Difference schemes
  - a) solves from 2 to jmax-1, 2 to kmax-1, 2 to lmax-1 (solve at interior points, apply boundary conditions at boundary points)
  - b) solves from 1 to jmax, 1, to kmax, 1 to lmax (solve at interior and boundary points)
- Solving at “cell-centers” with Finite-Volume schemes
- Use of C-grids, O-grids, etc.

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Personal chronology 1982-1983

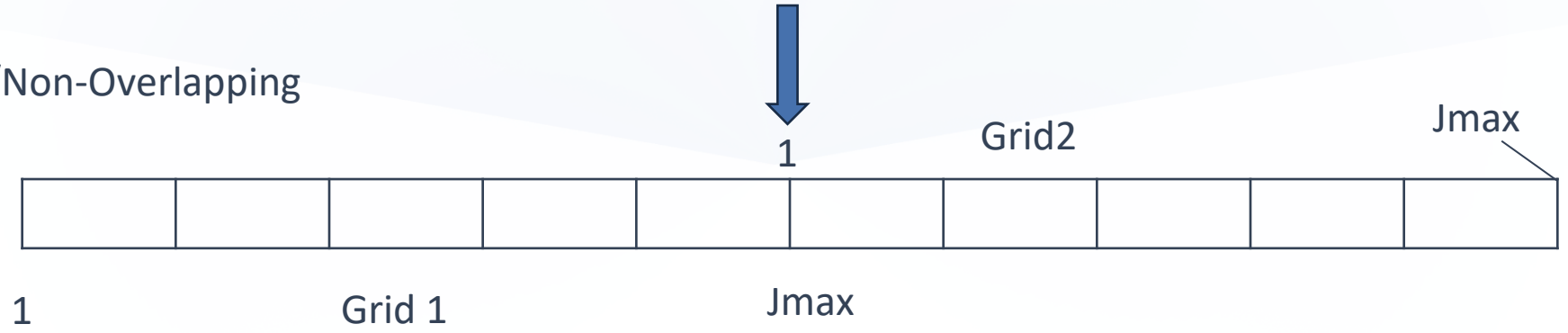
- Joe Steger at Stanford University, asked me to join him (as research associate) based on rapport built in the 70s
- Worked at Stanford Univ., on leave from Rockwell International Science Center
- Joe worked on central difference schemes, I worked on upwind schemes, as well as Incompressible Navier-Stokes solver using artificial compressibility
- Many discussions on how to handle complex geometries – how about multiple structured grid blocks instead of a single block
- Joe Steger went full steam ahead on overset composite grids, while I initially worked on ”patched” composite grids – multi-zone or multi-block
- I returned to Rockwell Science Center in 1983; eventually Joe left Stanford too

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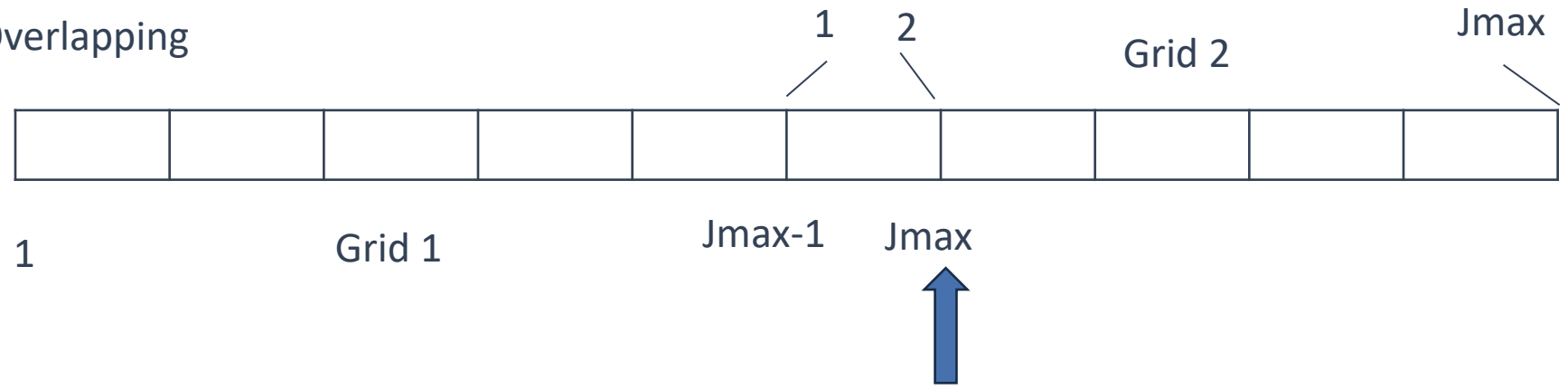
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Patched/Overset in 1-d

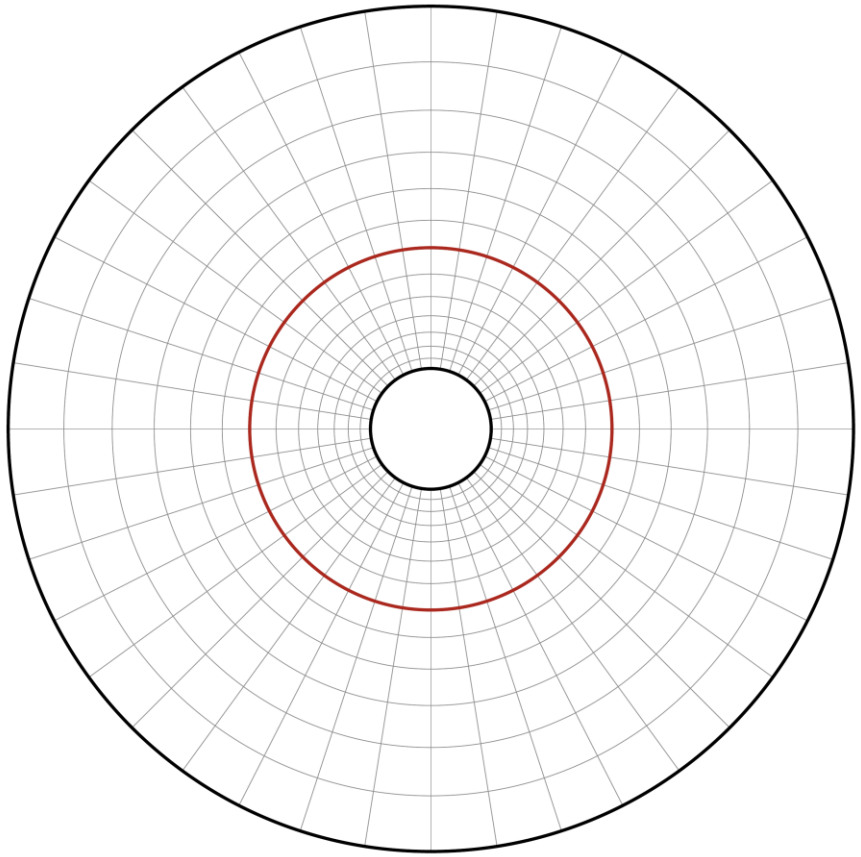
Patched/Non-Overlapping



Overset/Overlapping



Composite Grid Around Circular Cylinder



1982-83 First Composite Grid Application

Two grids – inner and outer

Inner grid – Euler equations

Outer grid – Potential flow equation

Proper boundary conditions at inter-block boundary

Foundation for later work

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Some transitions in the 1980s

- Finite-Difference schemes to Finite-Volume Schemes
- Differential conservation laws to Integral form of the conservation laws
- Difference representations to Polynomial representations
- Approximate Riemann Solvers
- Upwind schemes - for inviscid flows, acceptance for the inviscid part of flows with viscous effects – no loss of accuracy
- Upwind schemes - first order to second order, TVD schemes, non-oscillatory interpolation, avoiding spurious numerical oscillations
- Linear schemes for linear equations -> Nonlinear schemes for even linear equations
- Upwind schemes - Cartesian coordinates to curvilinear coordinates
- Implicit relaxation methods for Upwind schemes
- Leading up to Unstructured grid treatments in the 1990s

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Personal chronology - 1980s

Personal Chronology mid to late 80s
INS3D - Incompressible Navier-Stokes equations using Artificial Compressibility
EMTAC - Euler Marching Technique for Accurate Computation - Euler equations
EMTAC - MZ (Multi-Zone) - patched grid multi-block grids

Personal Chronology mid to late 80s
USA - Series codes - Unified Solution Algorithms
Unification of time-marching and space-marching
USA - PG for Perfect Gas - 2D, 3D
USA - RG for Real Gas - 2D, 3D
USA - EA for Equilibrium Air (Tannehill curve fitting of equations of state) - 2D, 3D
USA - IN for Incompressible Flows - 2D, 3D - Artificial Compressibility
Used for Space Shuttle and NASP applications

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Note on USA-Series Codes

- Solved the RANS equations or Euler equation subset
- Included shock fitting at boundaries, shock capturing everywhere (for compressible flow solvers)
- All editions Included structured multi-block overset grids as well as patched grids
- Overset grid algorithm evolved into being based on geometric cutting and blanking
- All “mesh processing” operations were performed within the flow solver (run time)
- All editions used explicit schemes or implicit schemes (approximate factorization) within each block – implicit schemes were the workhorse
- Vectorization to make use of vector computers available and popular and powerful then
- Sequential geometric multi-grid for some convergence acceleration
- Shift to completely dynamic memory allocation in C-language routines (solvers could handle all size problems without recompilation of the code)
- Extensive set of boundary conditions

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Personal chronology – early 90s

Universe Series - 1D, 2D, 3D rolled into one - All physics to be covered in one code framework and implementation

Formal development of higher than second-order treatments in polynomials and numerical fluxes

Formally based on the Integral law form of the conservation laws

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Personal chronology – mid 90s onwards

CFD++

Unified Grid, Unified Physics, Unified Computing

- Each cell could be any type, 1D, 2D, 3D
- Each cell could be assigned a different equation set
- Any grid point could have grid point velocity
- Unstructured and structured grids (digested as unstructured)
- Single unstructured grid can cover any complex geometry domain
- Multiple (composite unstructured) grids to cover multiple physics (FSI, FTSI, CHT), relative motion between meshes, mesh substitution for design campaigns
- Same piecewise-polynomial scheme at interior cells, boundary cells, conformal grids, non-conformal grids
- Same relaxation + algebraic multi-grid treatment everywhere (+ many other explicit implicit-explicit and other special schemes)

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Continuous Evolution of CFD++

Addition of pre-conditioning for certain types of flows
Addition of pressure-based algorithms
Addition of polyhedra and polygons
Addition of different physics in different mesh groups (e.g. Conjugate Heat Transfer)
More physics and even more physics
More ease of use
More speed
More integrated treatment with Metacomp's other software (CAA++, CSM++, METAFSI, MIME, SIM++)
Evolution of composite grid treatment – covered separately

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Continuous Evolution of CFD++

Single phase, multiple phases, phase transitions, cavitation, single species, multiple species, reactions (finite rate chemistry and equilibrium chemistry), thermal equilibrium and non-equilibrium, various equations of state - coverage for incompressible fluids to compressible fluids, ideal gas equations of state and cubic equations of state, radiation models, lots of turbulence models, lots of models for lots of physical situations (e.g. rotor/propeller modeling, porous region modeling, various source terms)

All applicable to Overset and Patched (aligned and non-aligned) steady and transient grid motions

All “mesh processing” operations carried out, even in multi-CPU mode, within the solver, with work distributed across the CPUs (MPI Ranks)

Mesh morphing

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Evolution of Overset Mesh Treatment in CFD++

- Cutting (with cutters) + blanking (to remove unwanted cells) + retention or deletion of the actual cut cells + definition of “live cells”
- Single rank -> Multiple ranks (Sequential cutting) -> automated cutting and blanking
- Whittling away at conservation errors at “zonal boundaries” (interface/ interaction) between the composite grids) - “flux-stitching”
- Many Applications over the years
- Ease of use improvements and efficiency improvements
- Future - More automation, more efficiency, more accuracy, more physics, more ease of use and effectiveness overall

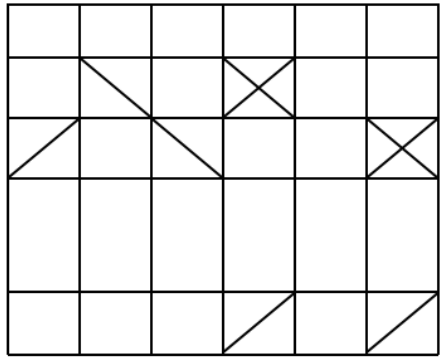
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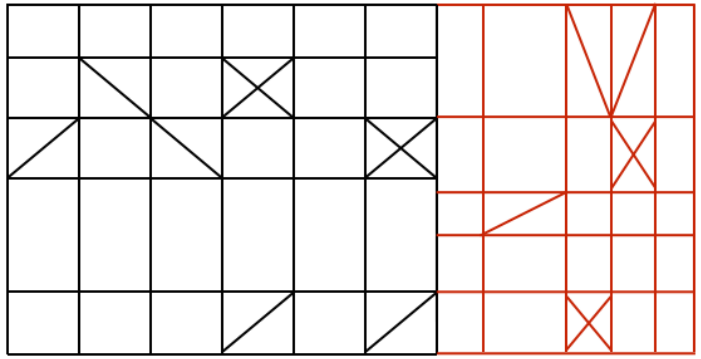
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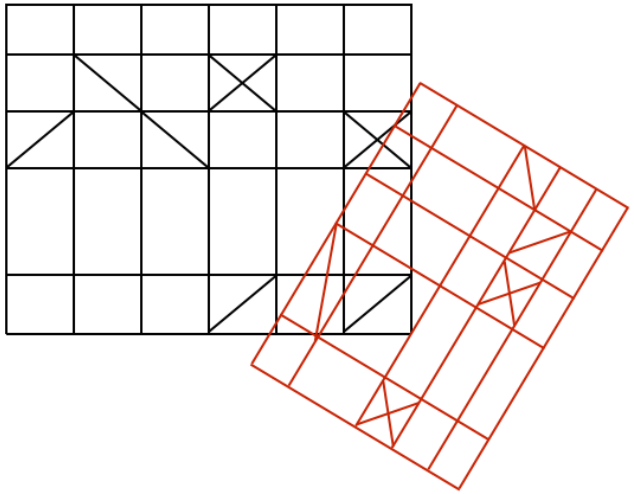
CFD++ Grid Types

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CFD++ Grid Types

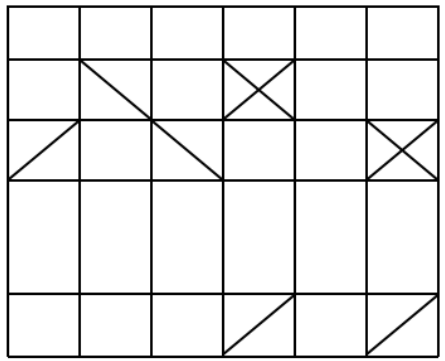


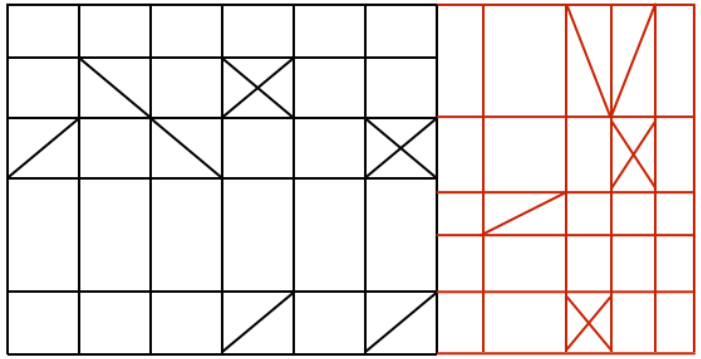


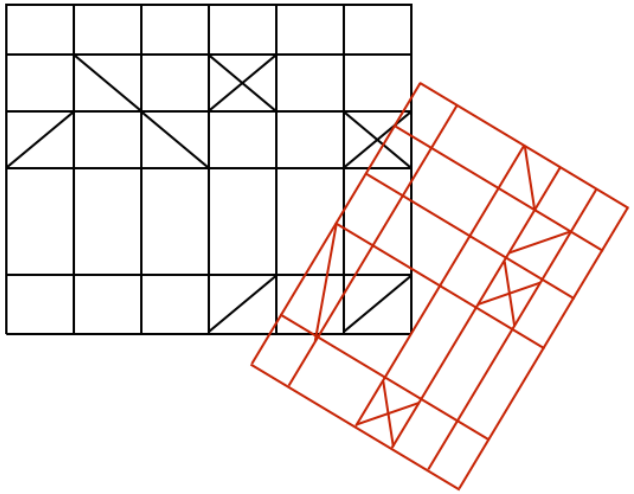


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CFD++ Grid Types





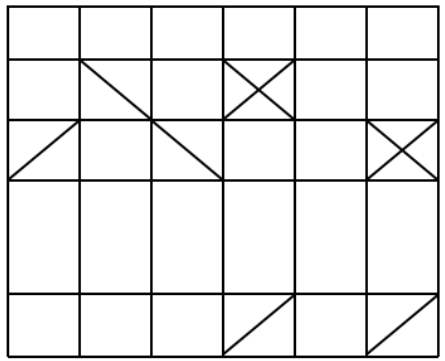


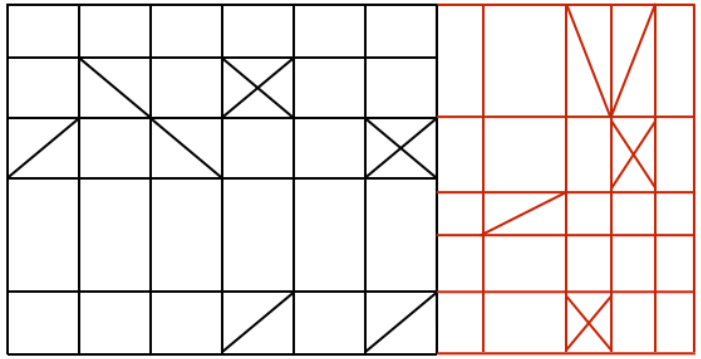
Conformal grids are those that are made up of cells, each of which have faces that have exactly one neighboring cell.

No special boundary or cutting/blanking treatment needed.

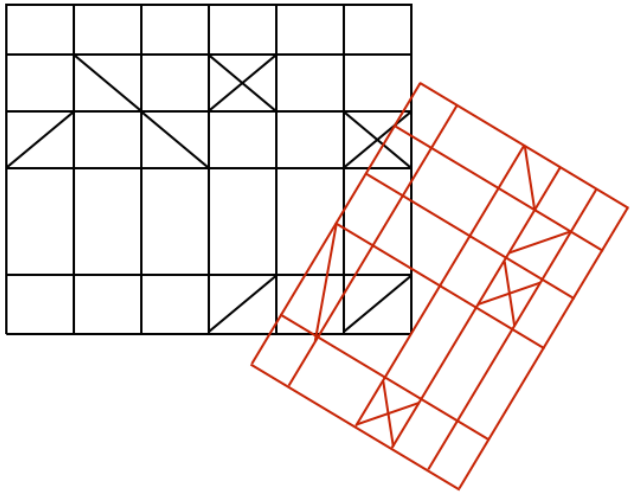
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CFD++ Grid Types







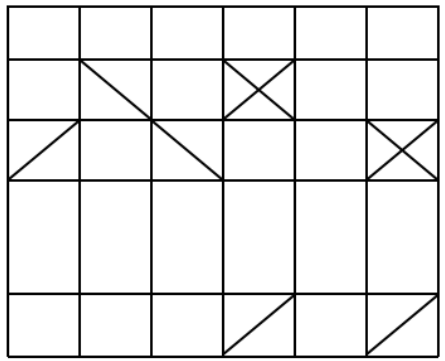


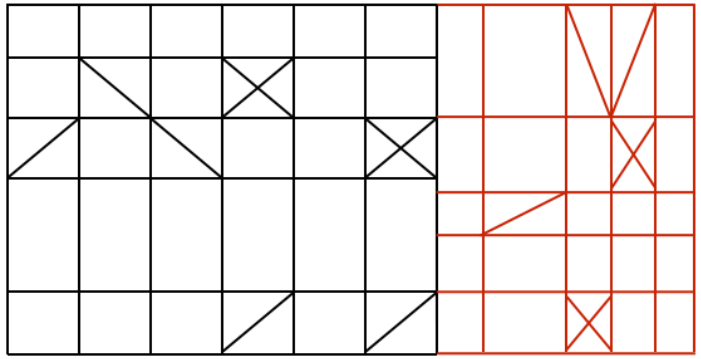
Non-conformal patched non-aligned grids are usually divided into different regions called “cell groups”

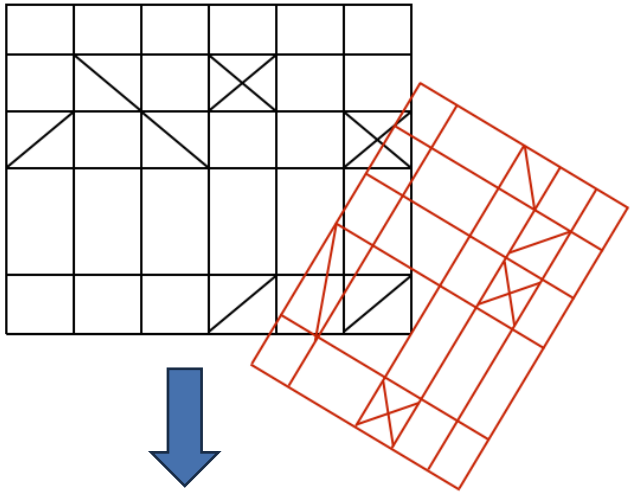
Groups are connected via zonal boundaries that have no overlap. Cutting/blanking procedure not needed.

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CFD++ Grid Types





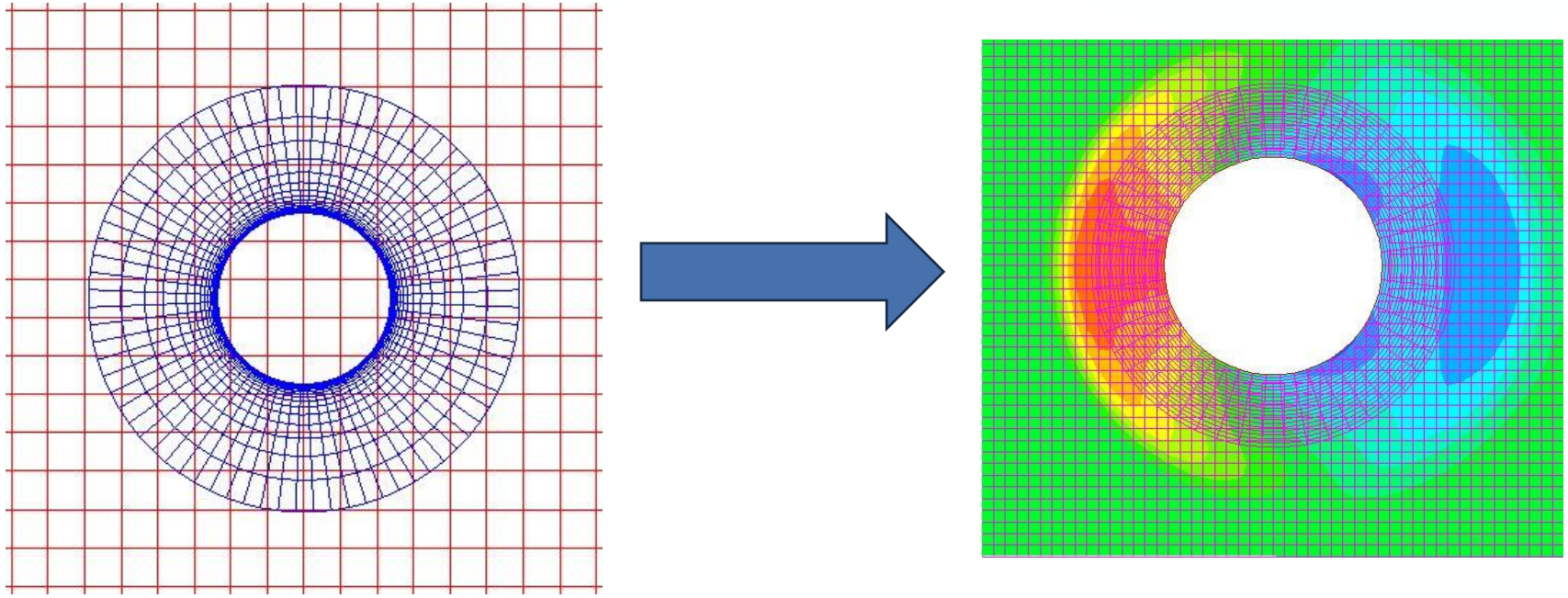


Non-conformal, overset grids may require cutting/blanking procedure to create proper overlap between cell groups.

A zonal boundary condition will also be needed for different groups to exchange flux information .

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Overset



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CFD++ Grid Motion

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Moving Meshes and Morphing

Moving meshes:

- Unique capabilities for steady and unsteady flows over complex geometries including bodies in relative motion
- Sliding and overset meshes
- Accurate treatment of conservation for such meshes
- Sequential cutting approach for cutting & blanking
- Global and body frame motion modes
- Integrated rigid body dynamics (RBD) capability with a six-degree-of-freedom (6DOF) module
- Co-simulation possibilities in 6DOF mode

Mesh morphing:

- RBF-based mesh morphing available via tool and solver (transient mode)
- File-based and BC-based mesh morphing modes
- Automatic periodicity of motion
- Multi-CPU mesh morphing (tool and solver)

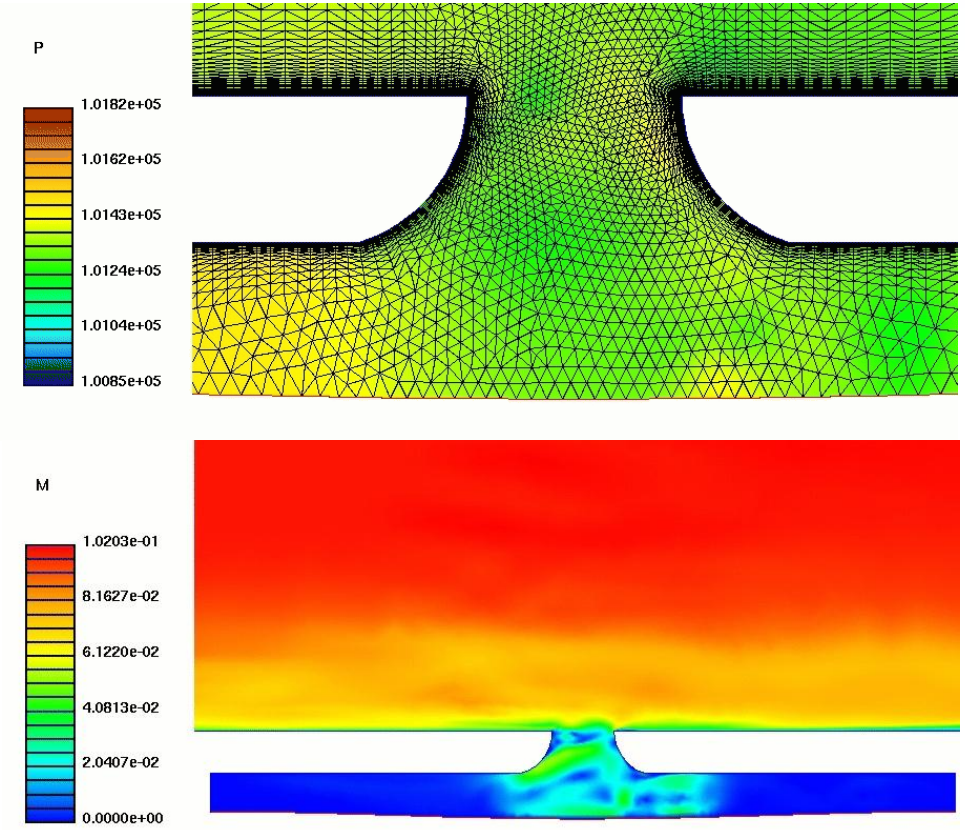
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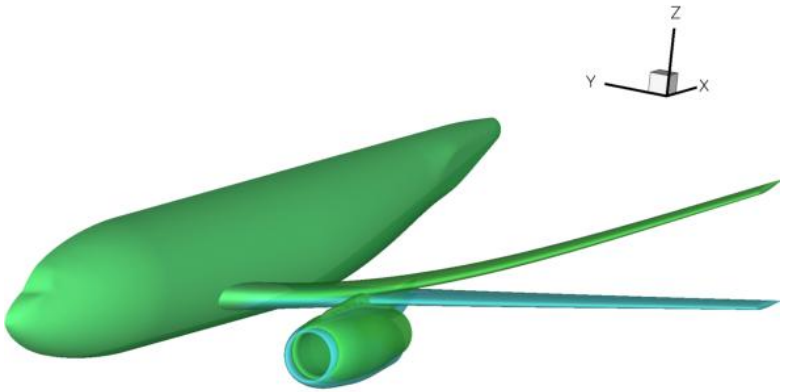
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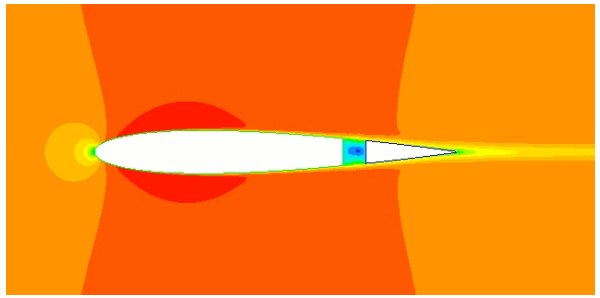
Composite Grid Applications

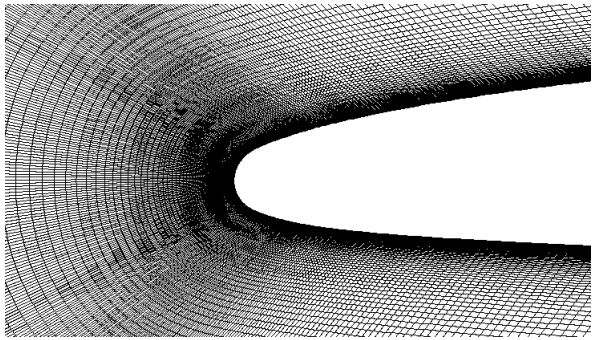
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Mesh Morphing (No Zonal BC Required)



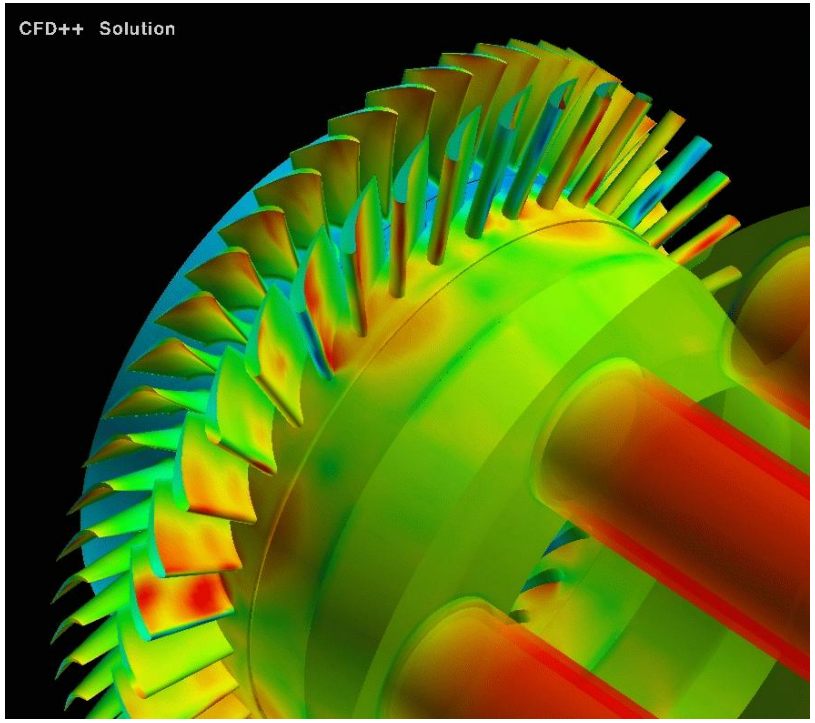


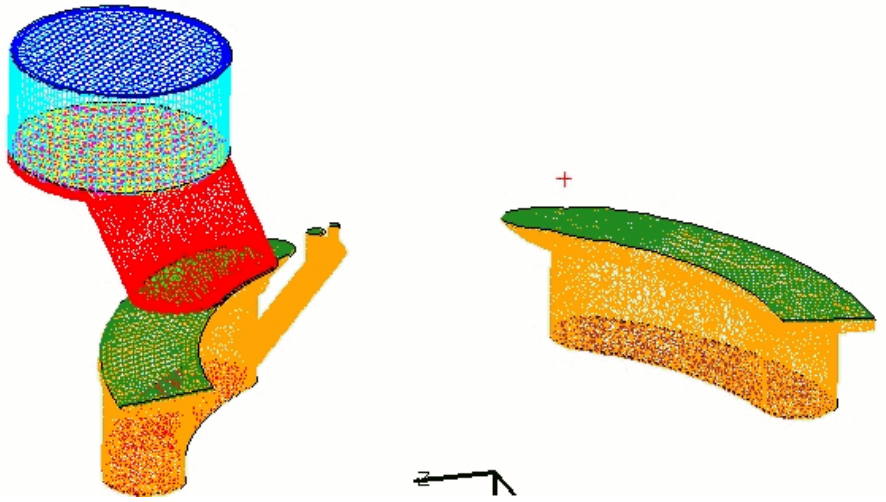




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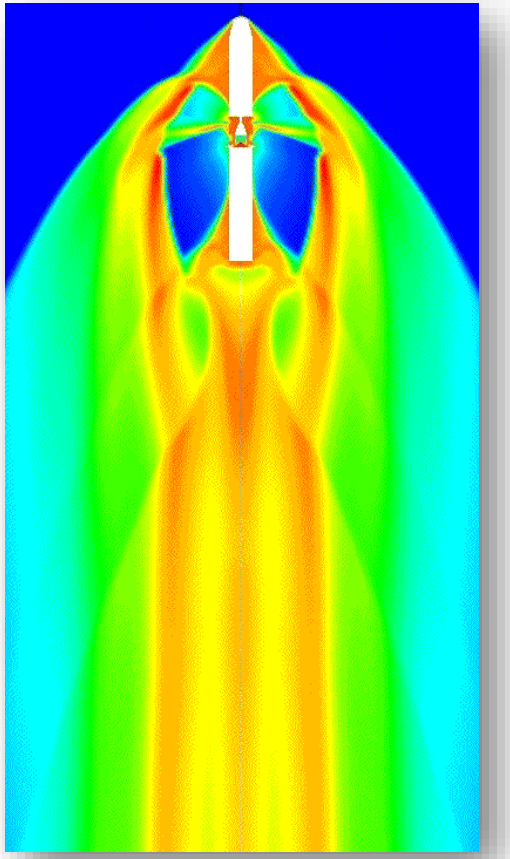
Sliding Mesh Motion (Patched Zonal)

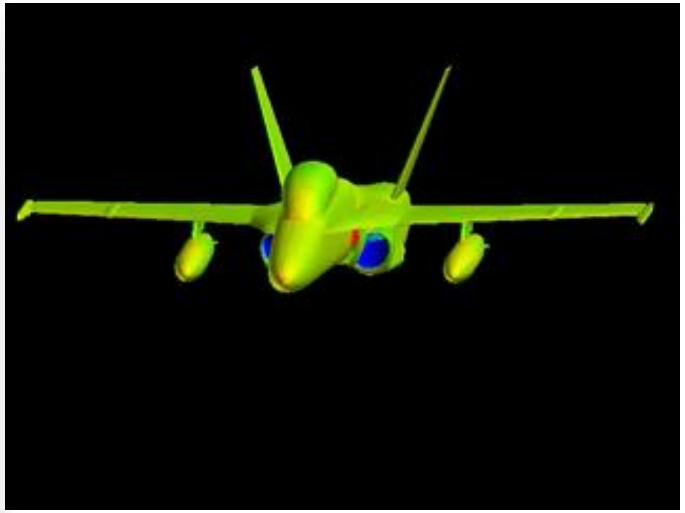


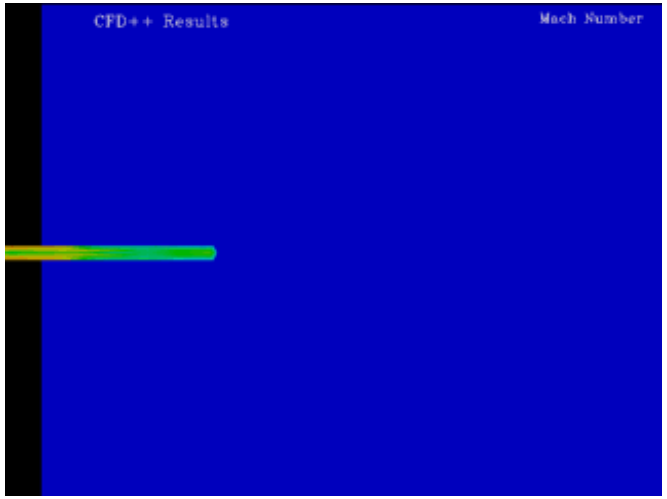


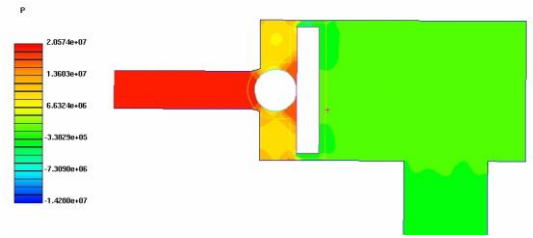
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Overset Mesh Motion (Patched and Overset Zonal)

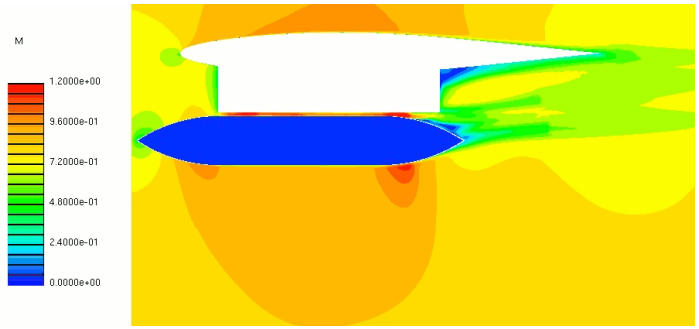


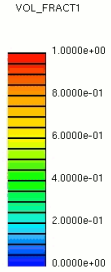


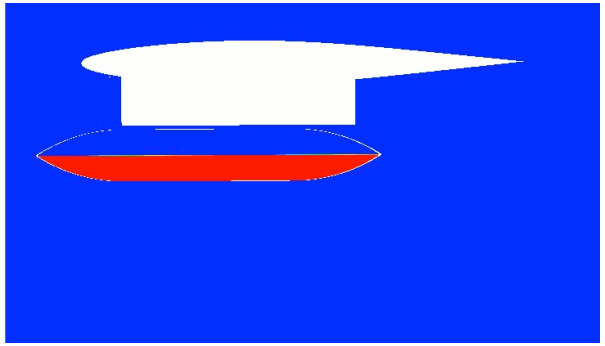








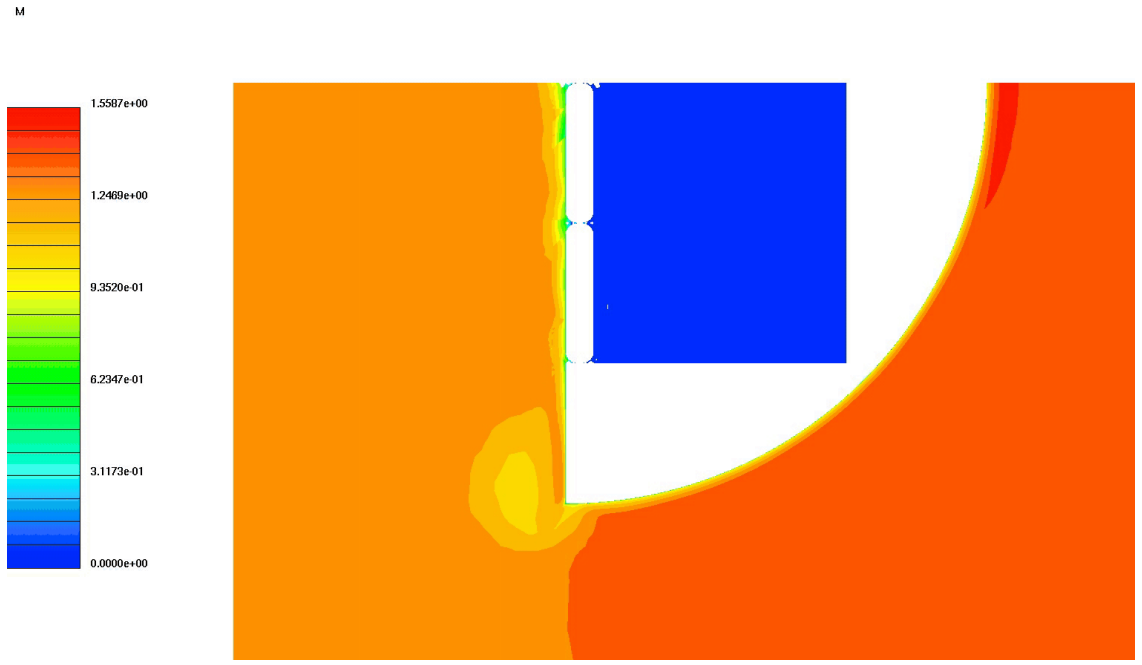




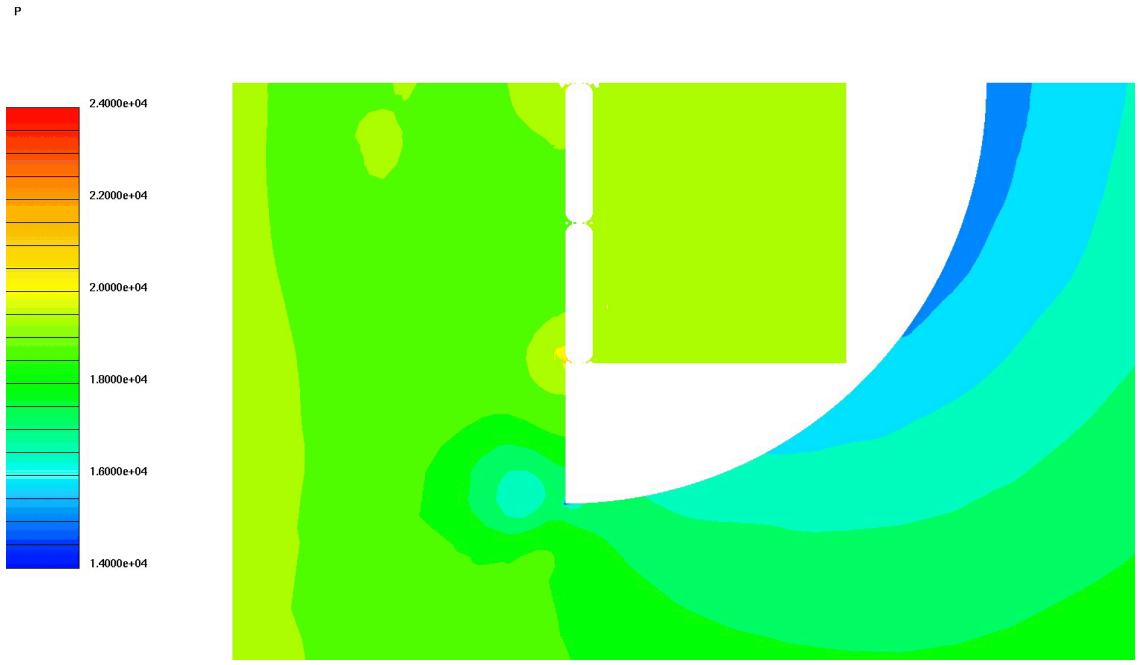
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Weapon bay door opening

M = 1.3, altitude = 12 km



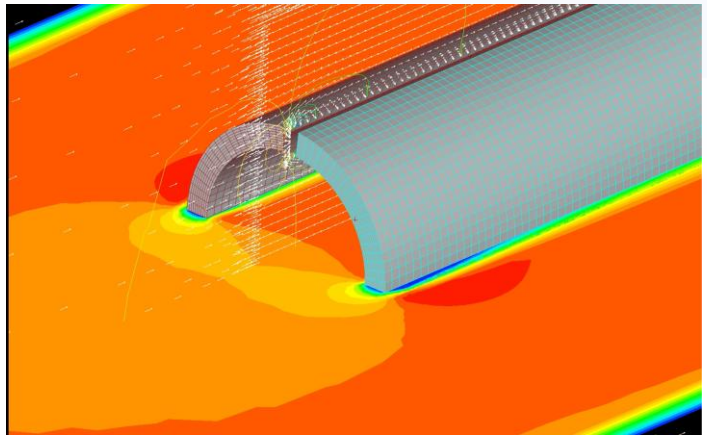
Evolution of Mach number



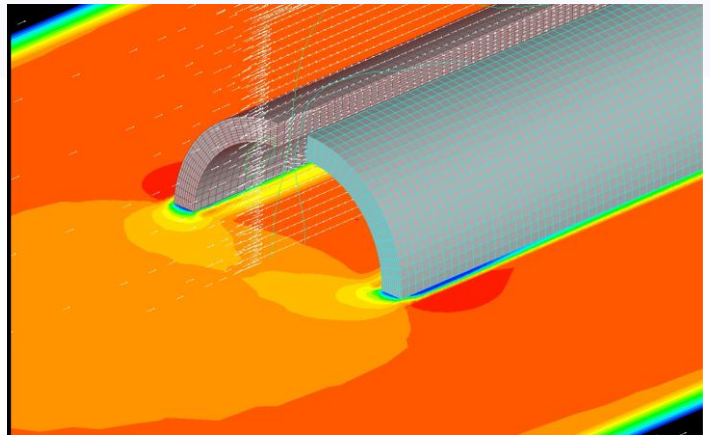
Evolution of Pressure

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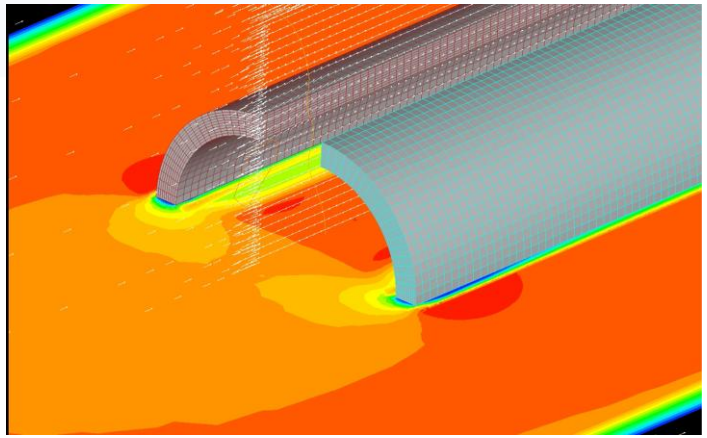
Semi-cylindrical shells disengaging in an incompressible flow through a duct



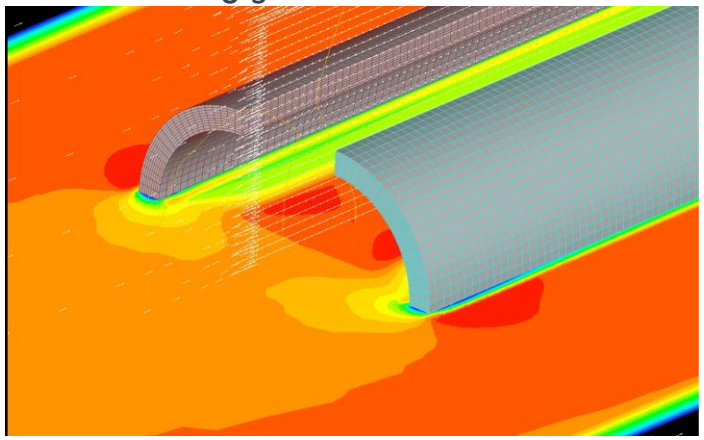
0 s



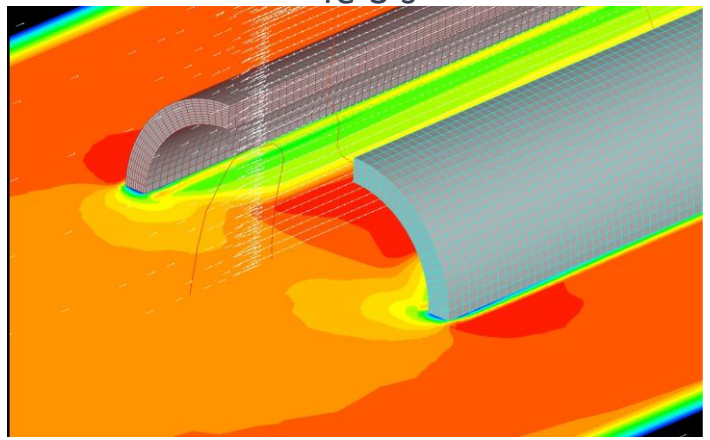
4e-3 s



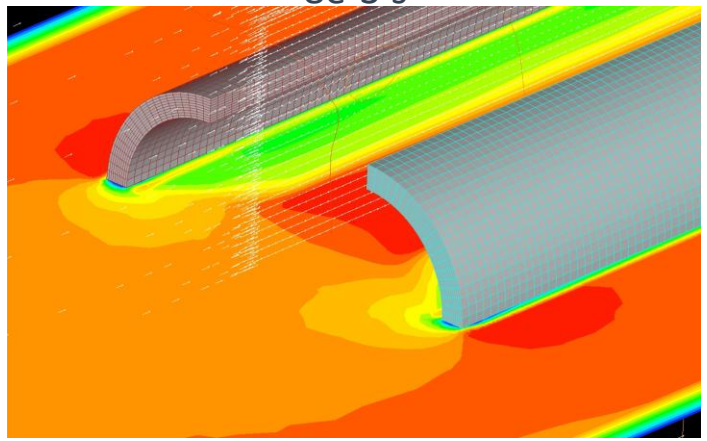
8e-3 s



12e-3 s



16e-3 s



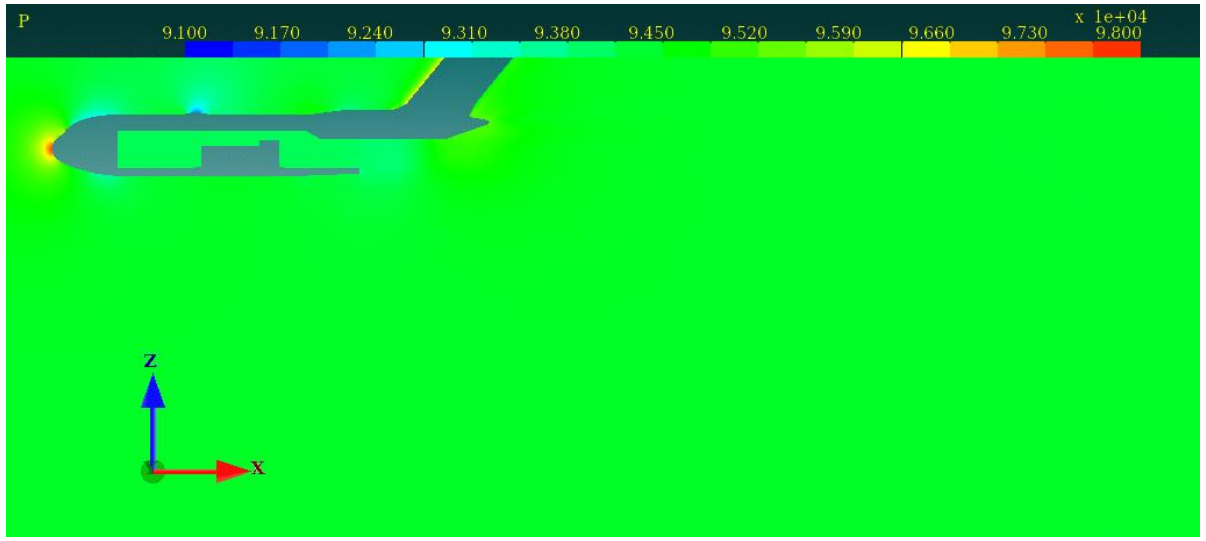
20e-3 s

The solid surface of the shells overlaid with its surface mesh, axial velocity contour field in the mid-y plane, pressure contour field (shown as contour lines) and velocity vectors in mid-z plane are shown.

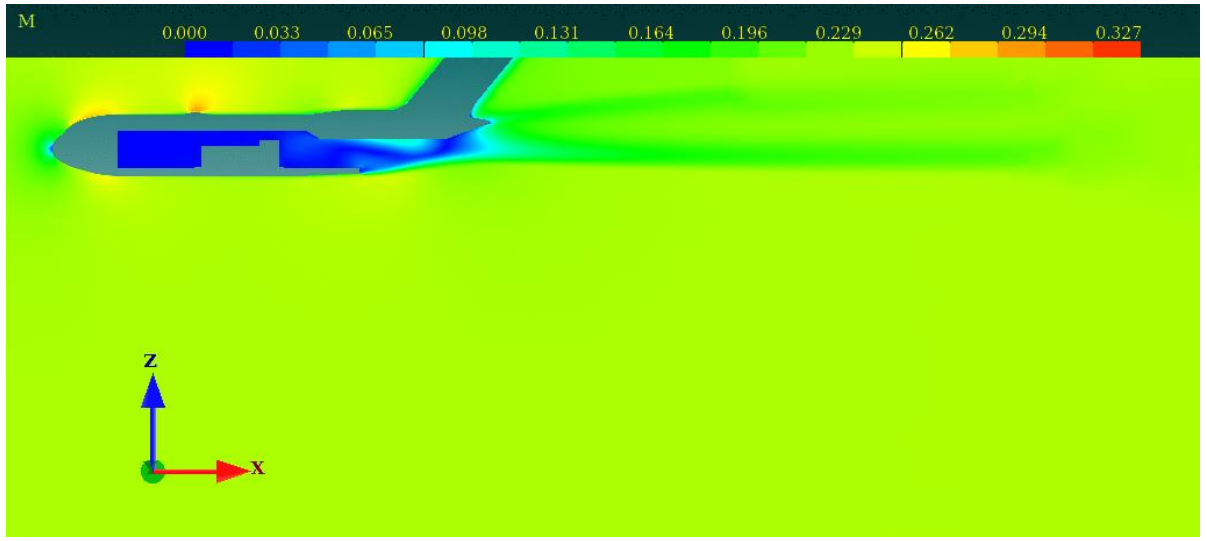
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Store drop from aircraft - 3D

- 3D case, 1DOF motion till aircraft exit with extraction force as a function of time, then 6 DOF motion due to gravity (M = 0.2, altitude = 0.6 km)



Evolution of Pressure at the mid-span plane

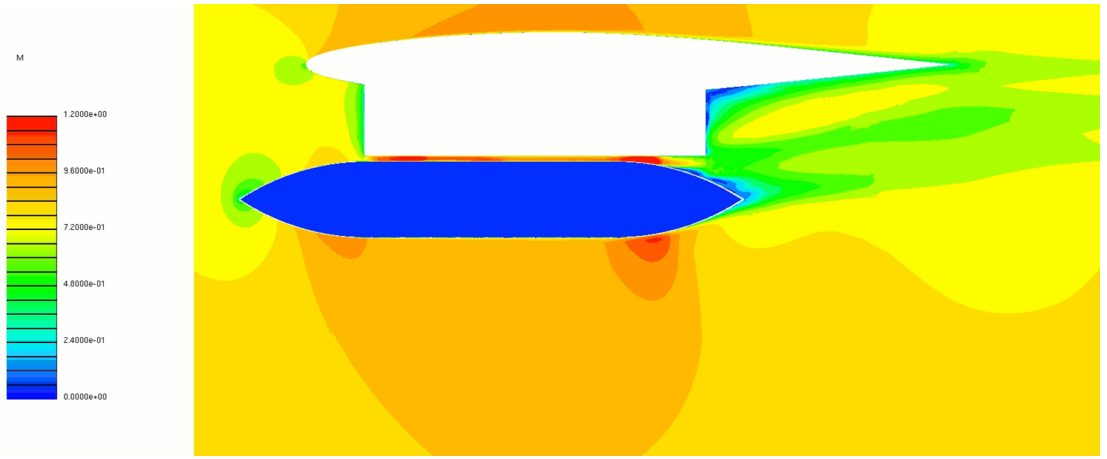


Evolution of Mach number at the mid-span plane

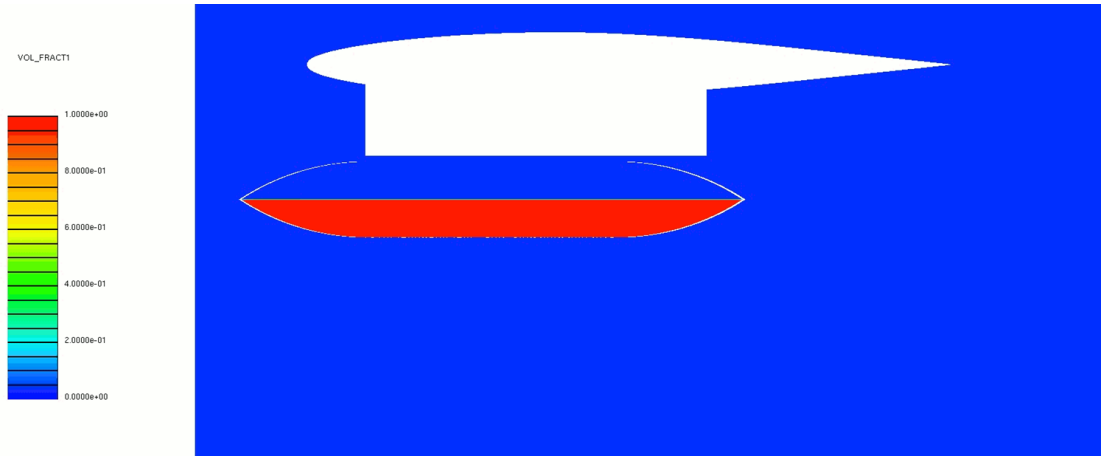
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Partially filled drop tank falling from moving aircraft wing

M = 0.8 flow over a partially filled fuel tank is simulated (altitude = 5 km).
Motion is prescribed for the fuel tank to simulate its drop.



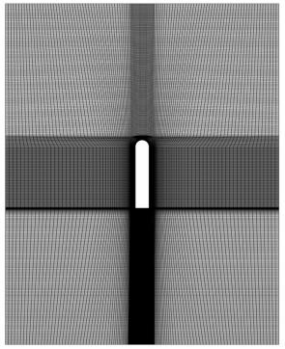
Evolution of Mach number

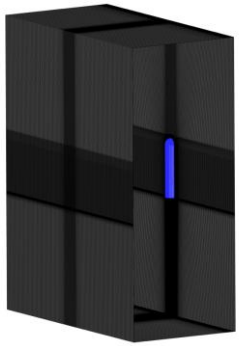


Evolution of Volume Fraction contours

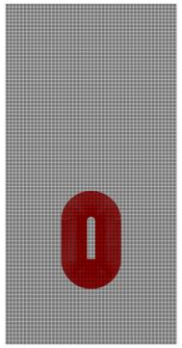
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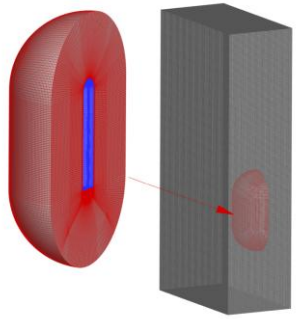
Motion of Cavitating Submerged Projectile



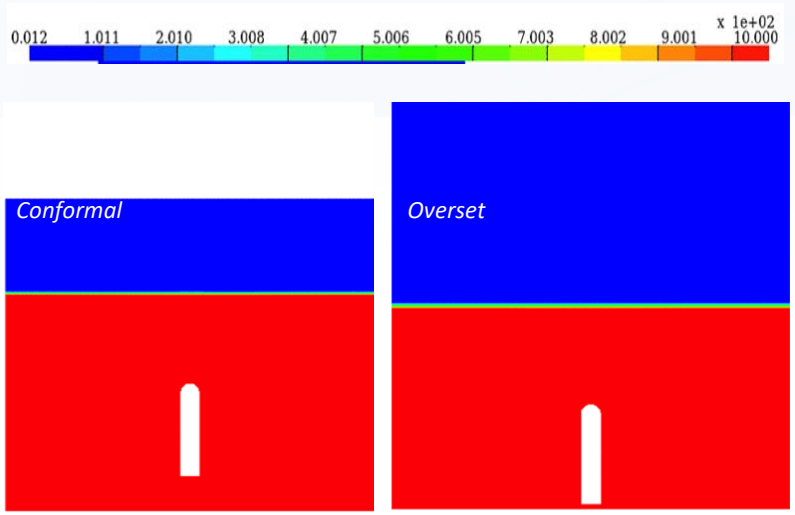


Conformal Single Moving Domain

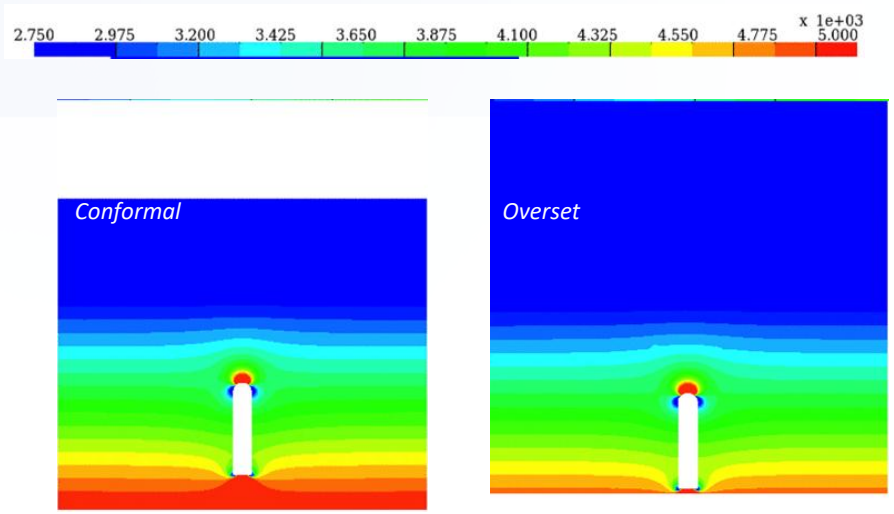




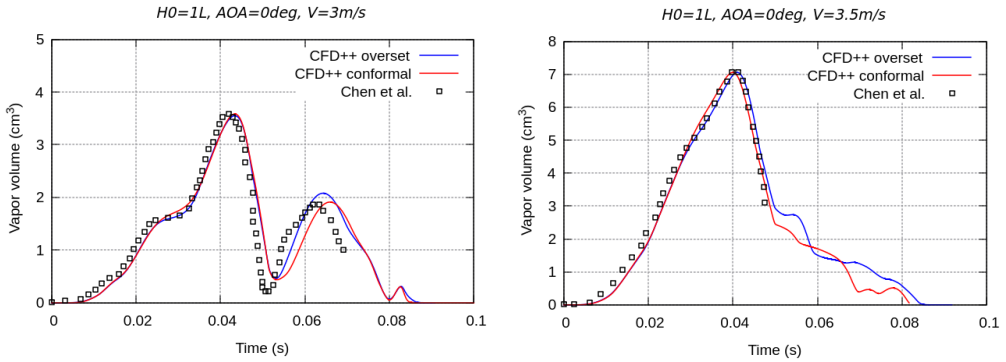
Overset



Density at Symmetry Plane



Pressure between 2750 Pa and 5000 Pa



Time history of total water vapor volume

Ref: Ying Chen, Jie Li, Zhaoxin Gong, Xin Chen and Chuanjing Lu, 2021, LES investigation on cavitating flow structures and loads of water-exiting submerged vehicles using a uniform filter of octree-based grids, Ocean Engg, 225, 108811

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Ricochet/Water Entry of Autonomous Underwater Vehicle

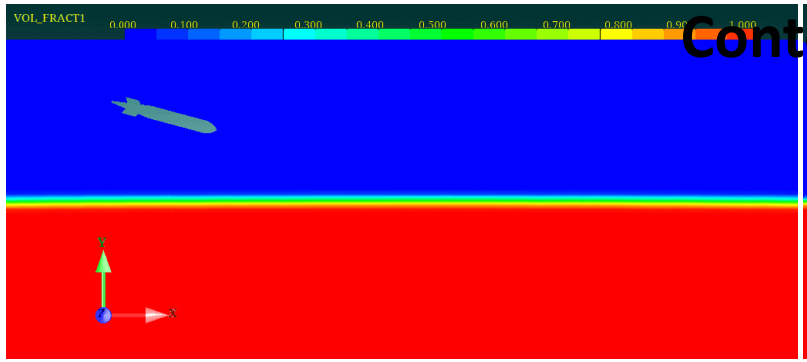


γ = 0°



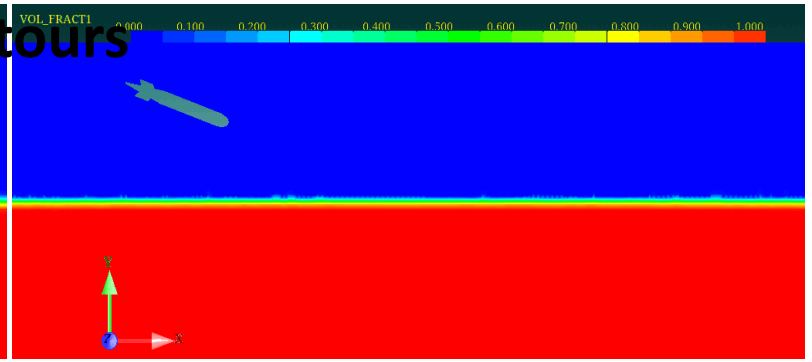
γ = 60°

Volume Fraction Contours



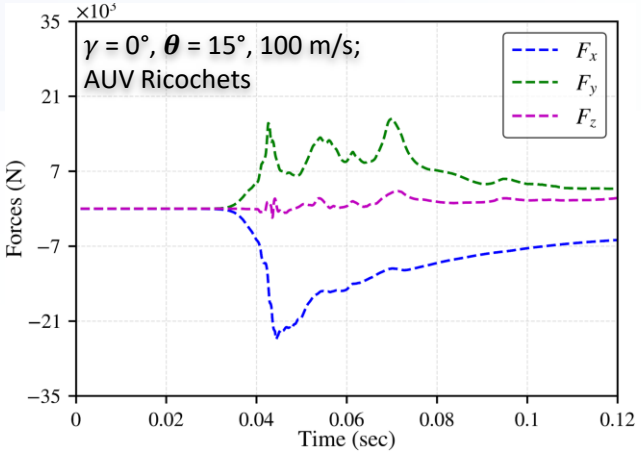
γ = 0°, θ = 15°, v = 100 m/s

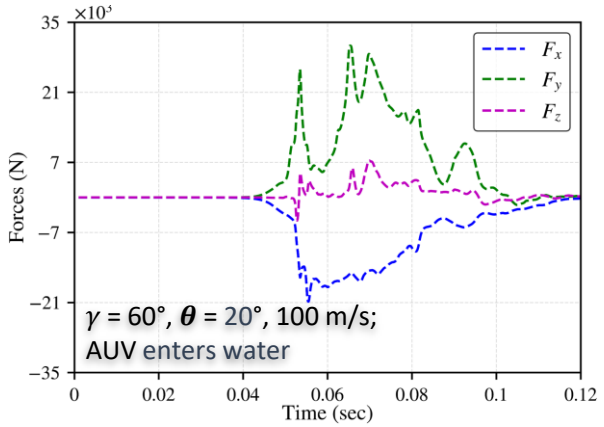
AUV Ricochets



γ = 60°, θ = 20°, v = 100 m/s

AUV enters water

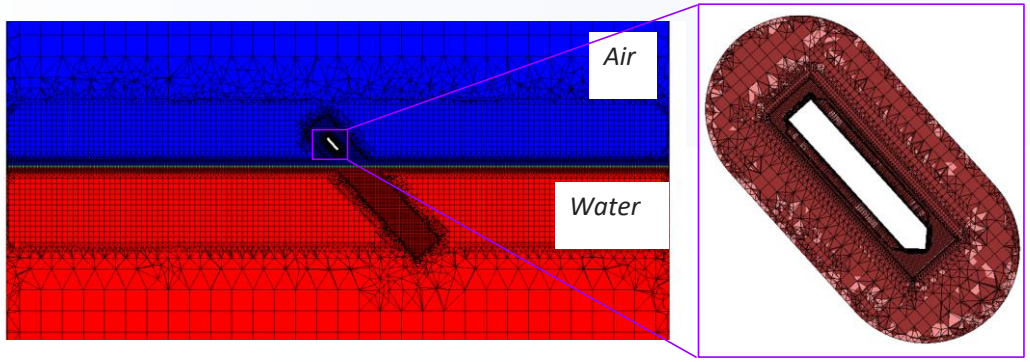


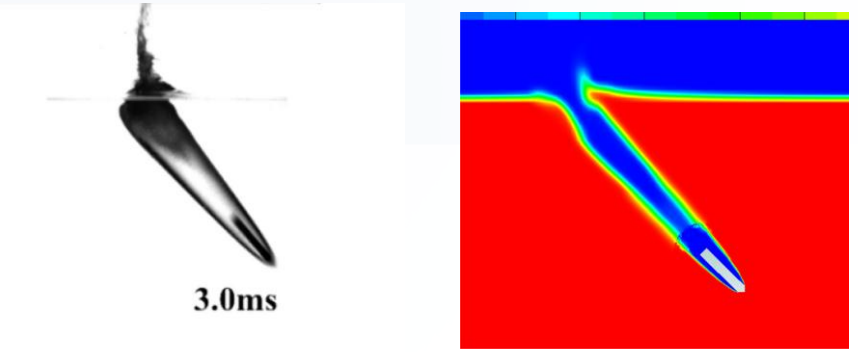


Ref: Guo-Xin Yan, Guang Pan and Yao Shi, Ricochet Characteristics of AUVs during Small-Angle Water Entry Process, Volume 2019, Article ID 9518437, Mathematical Problems in Engineering

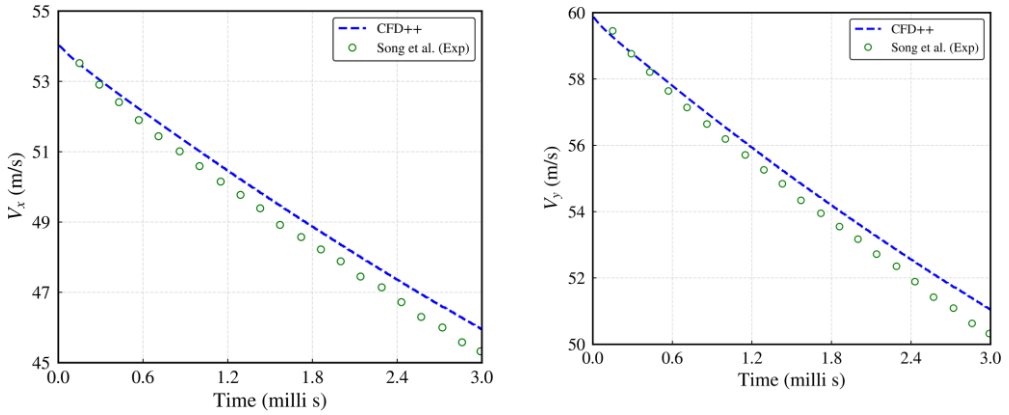
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Water Entry Dynamics of High-Speed Projectiles

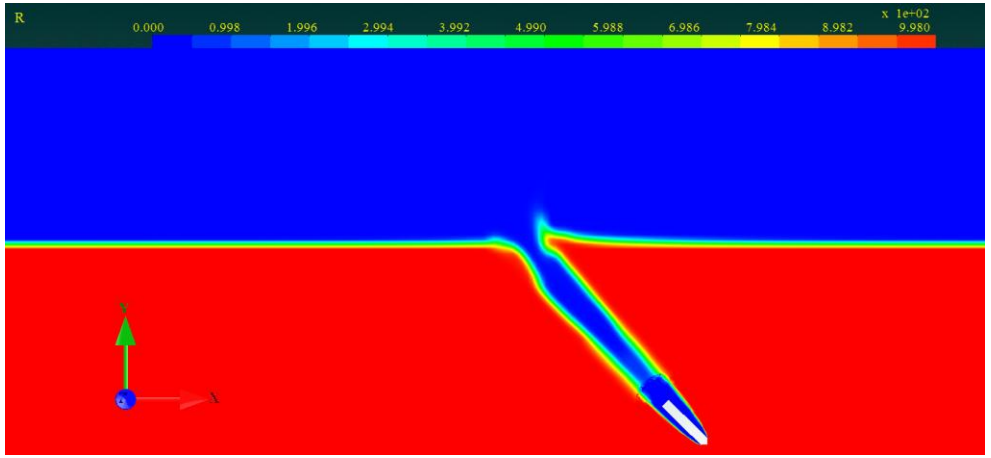




The cavity profiles by the oblique water entry
β = 48°, Vo = 80.088 m/s



Time history of velocity components

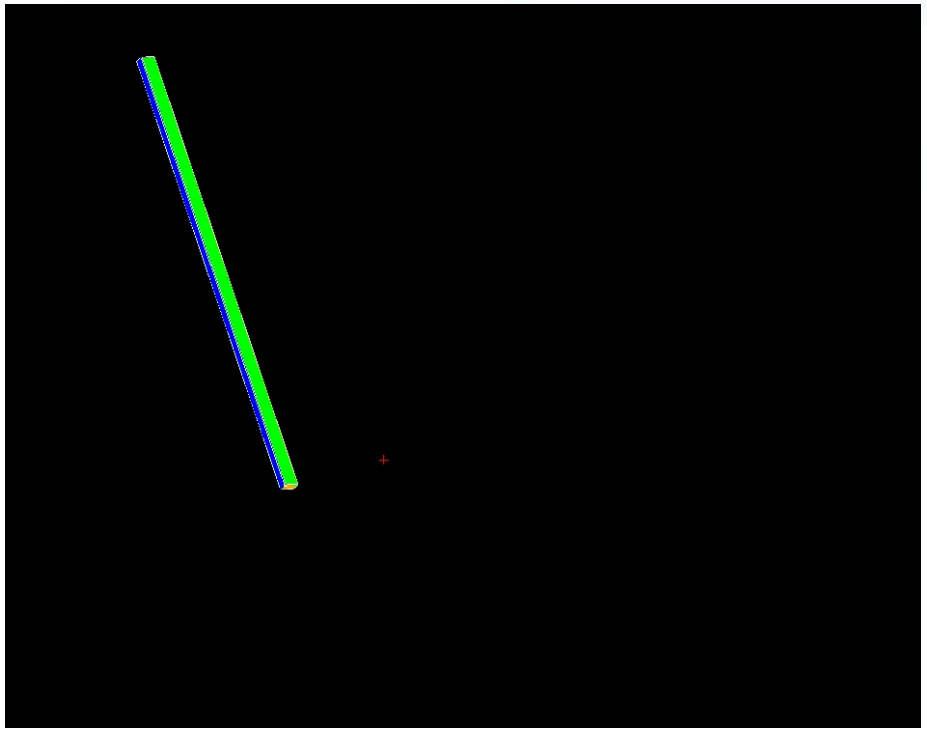


Density at a lateral cut plane

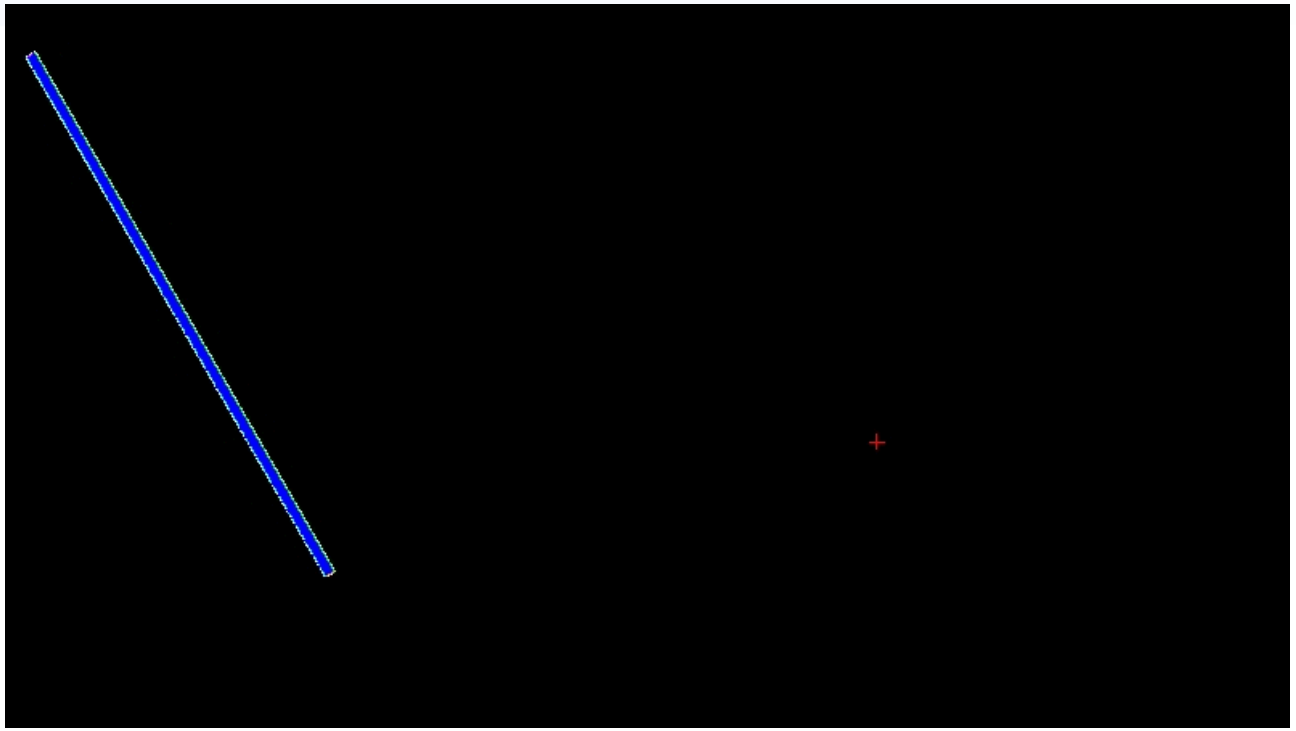
Ref: Song, Z. J., et al. "Experimental and numerical study of the water entry of projectiles at high oblique entry speed." Ocean Engg 211 (2020): 107574.

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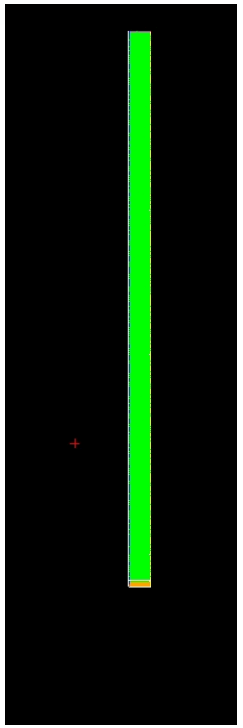
Modelling of Deforming Wiper Motion



Isometric view



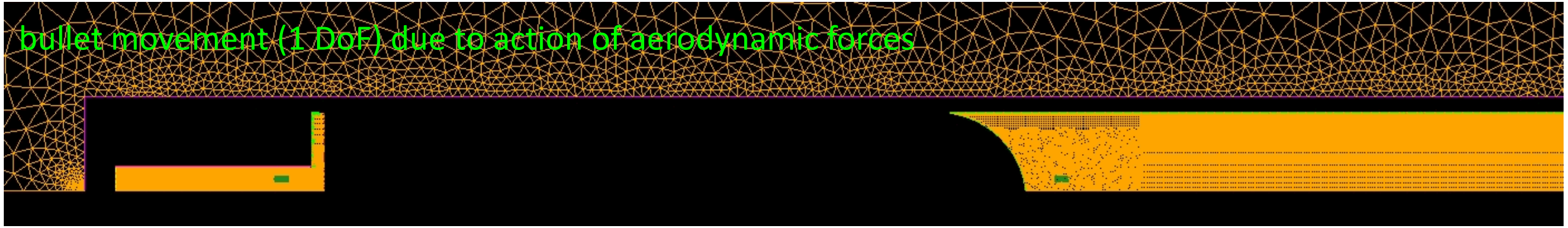
xy (top view)

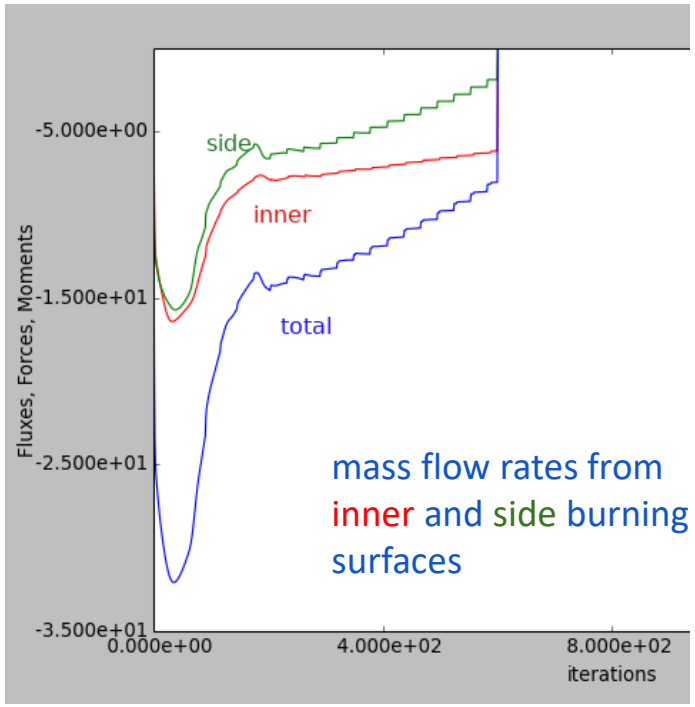


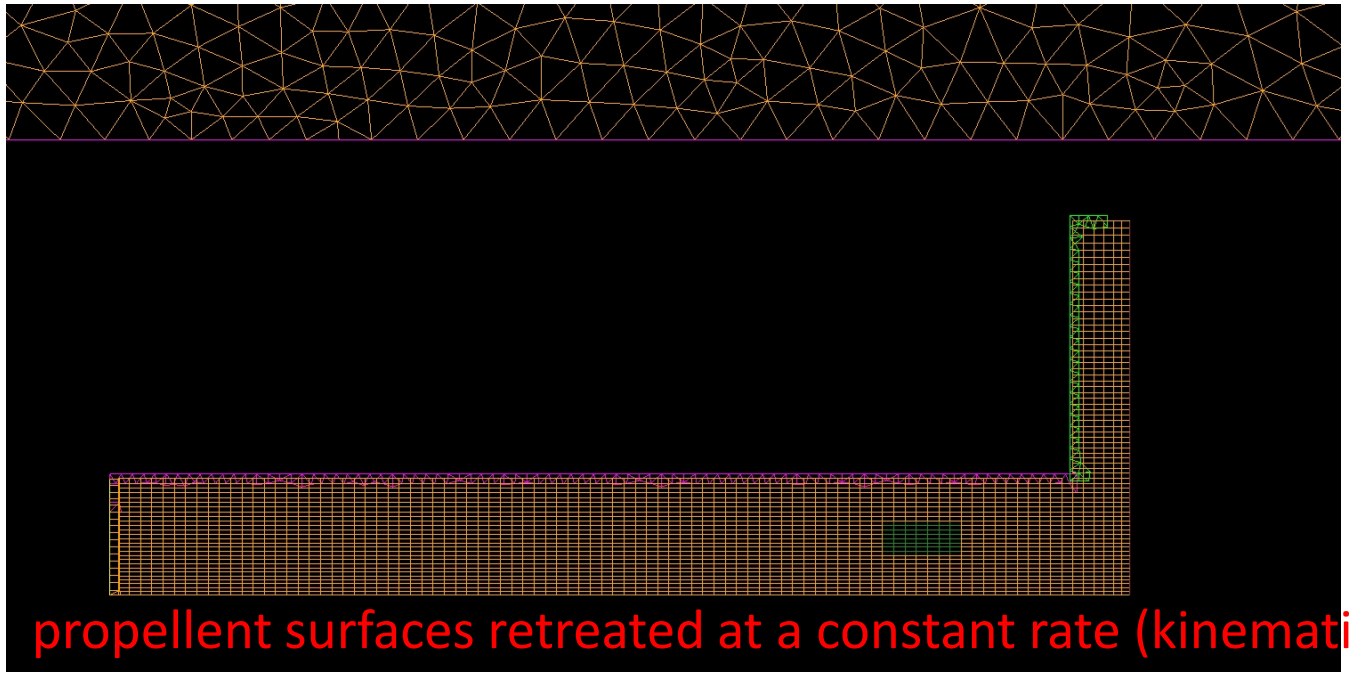
yz (side view)

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Propellent grain burning accompanied by bullet ejection from a barrel







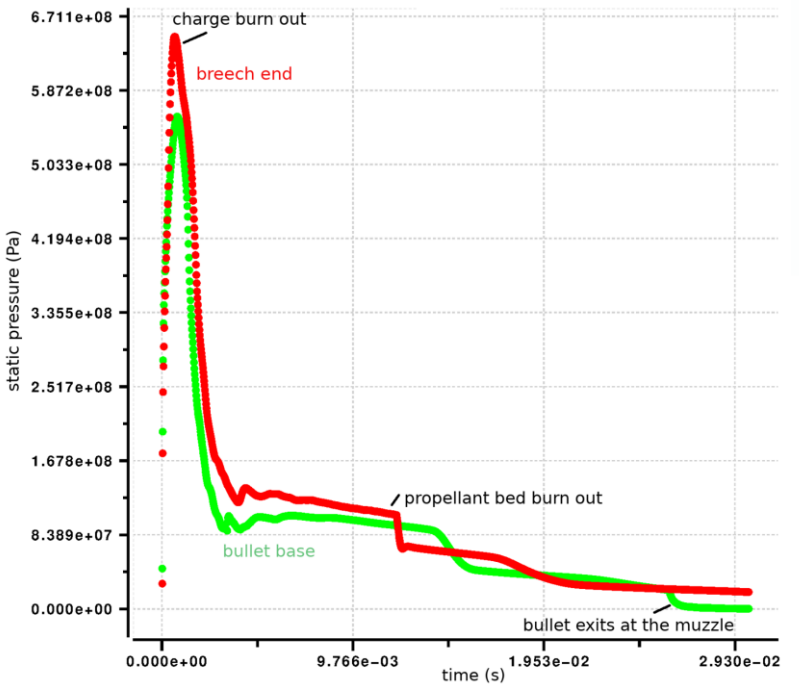
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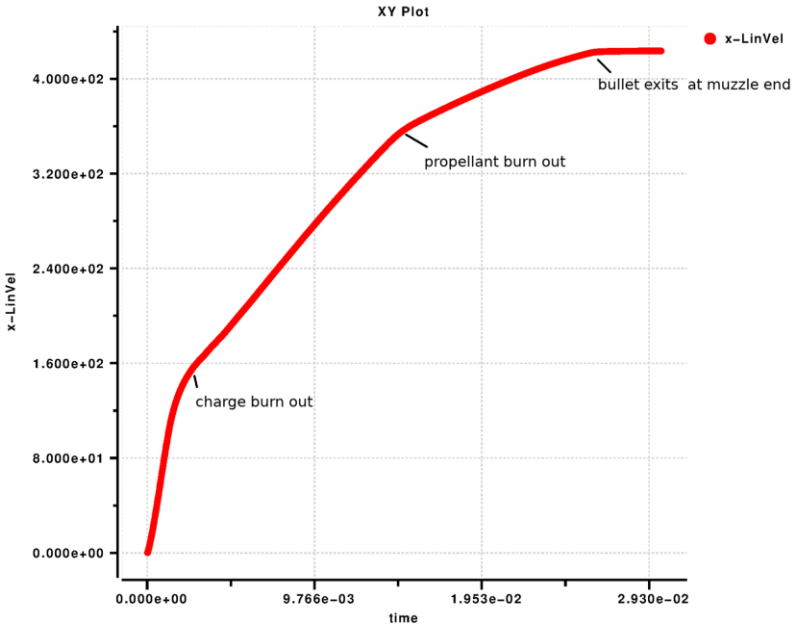
Propellent grain burning accompanied by bullet ejection from a barrel - Concluded





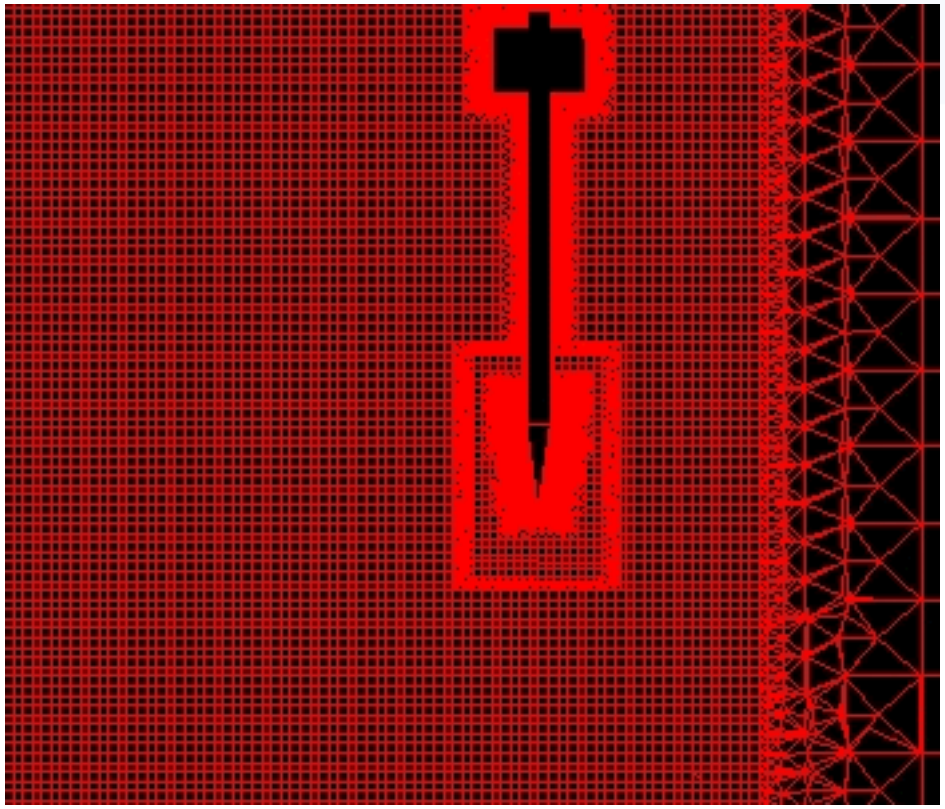


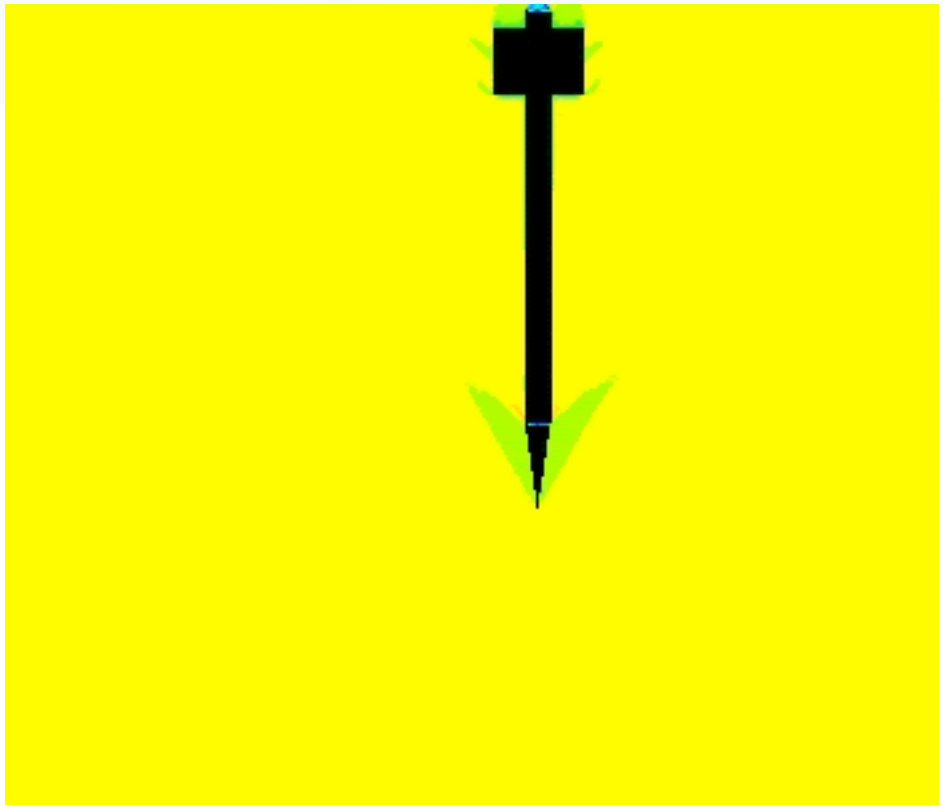




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Projectile fuse detachment

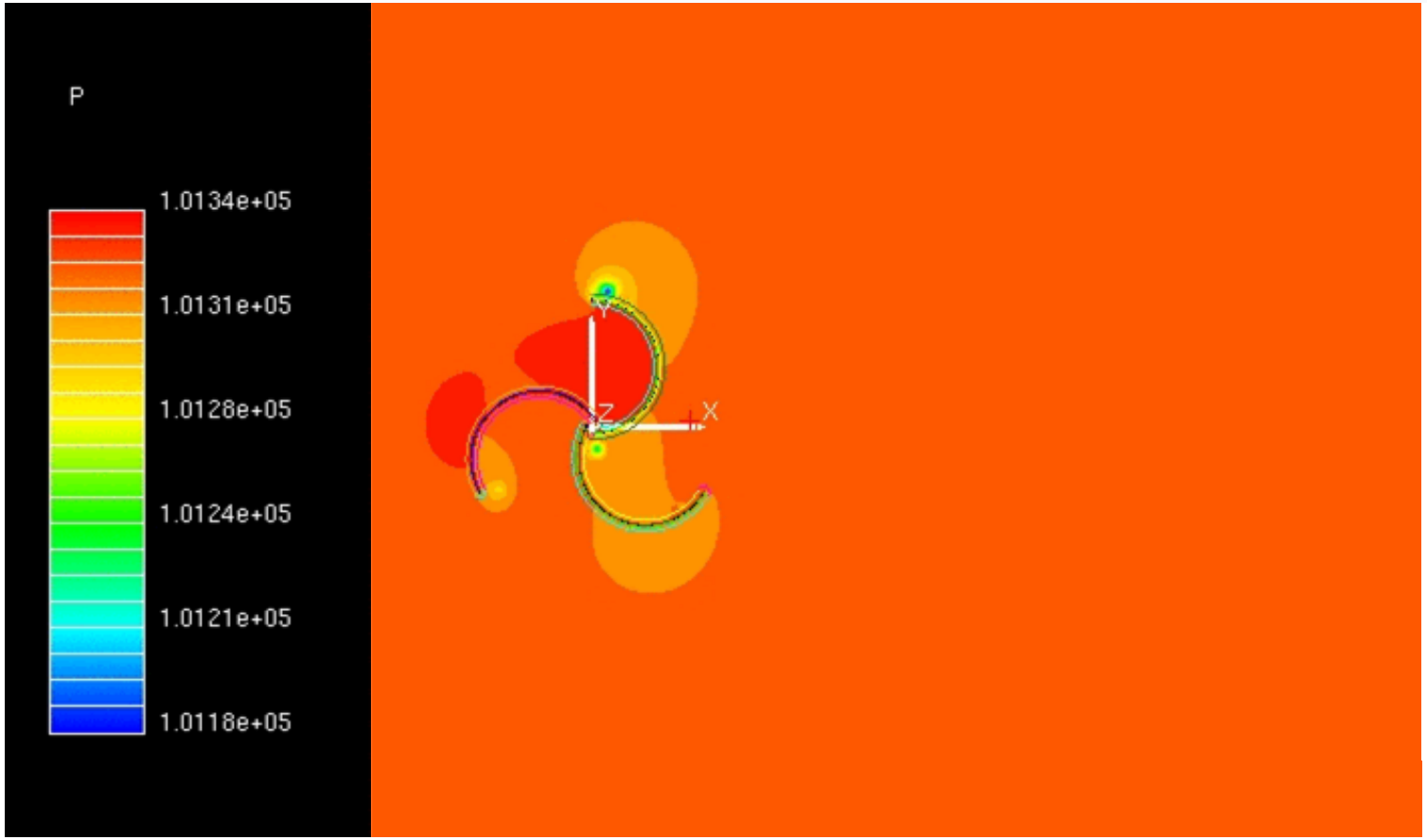


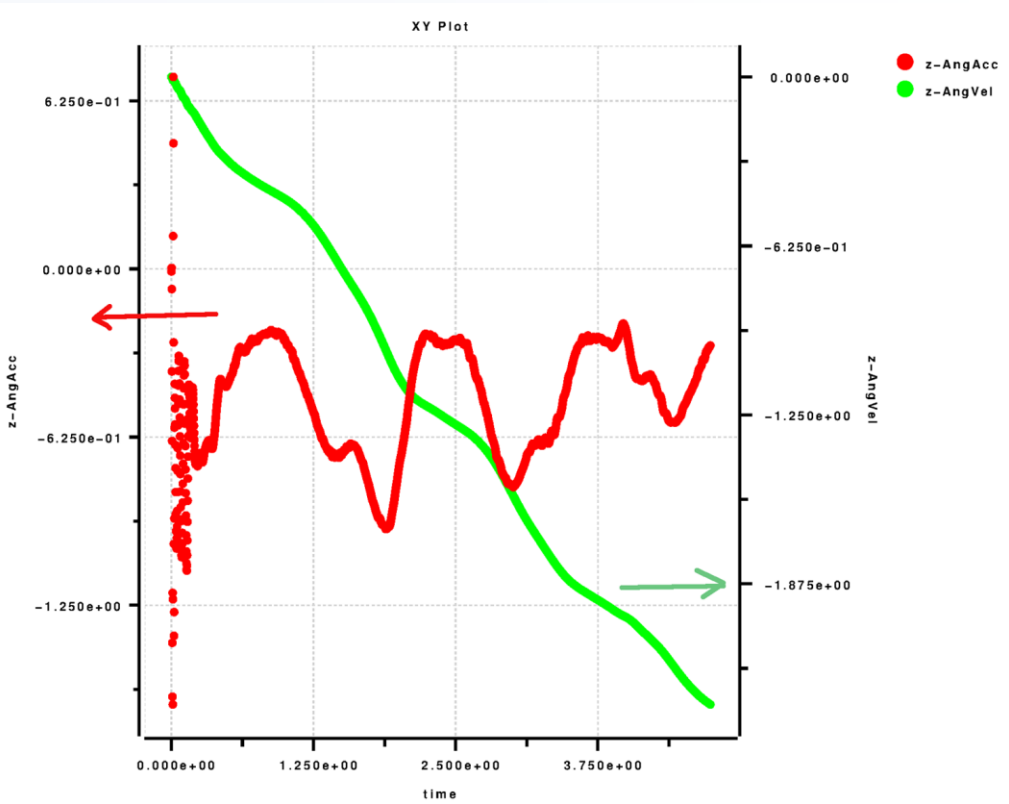


M = 1.3, altitude = 0.0 km

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Vertical axis wind turbine





u = 3 m/s at sea level

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Non-Concluding Remarks

I have been connected with overset composite grid solvers from 1982

My colleagues and I continue to have ideas on how to make things better

We hope to continue the evolution

Thank you

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